

BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science Engineering, Term Final (Fall) Examination 2022: Jan-Feb
2023

Student Group: Earned Credit Hours > 108

Subject: CSE 415, Human Computer Interaction

Time: 3.00 hours

Full Marks: 180

INSTRUCTIONS:

- a. Use **SEPARATE** answer scripts for each section.
- b. **Question-1** in Section-A and **Question-5** in Section-B are compulsory
- c. Answer any other **TWO** questions out of **THREE** from each section.
- d. Figures in the margin indicate full marks.
- e. Assume reasonable data if necessary.
- f. **Symbols and abbreviations** used have their usual meanings.

SECTION-A

Question 1 (Compulsory)

- a. Choose an appropriate evaluation method for the following situation: 3+18
You have designed and implemented a new game system and want to evaluate it before release,
Also, describe the following in brief:
i) the participants
ii) The technique used to evaluate
iii) Representative tasks to be examined
iv) An outline plan for carrying out the evaluation.
- b. Distinguish between the 'Cognitive Walkthrough' and 'Heuristic Evaluation.' 6
- c. State the factors that define the severity of a usability problem. 3

Question 2

- a. Evaluate the MIST website based on the seven general principles of Universal design.; 14
- b. Write short notes on the following concepts: 9
(i) Earcons
(ii) Multi-modal
(iii) Auditory icons
- c. State the primary activities of the DECIDE framework to guide evaluation. 3
- d. Define 'Persona' with suitable example. 4

Question 3

- a. Highlight the importance of 'Scenarios' for design. Provide two example scenarios considering the context of designing an e-commerce website for selling cosmetic items, 10
- b. Why is context important in selecting and applying guidelines and principles for interface design? Illustrate your answer with examples. 10

- (i) Multi-threading
- (ii) Task migratability.

Question 4

- a. Depict the Information Architecture (IA) of a museum website following the steps of 'Card Sorting' technique. At least 20 concepts need to be considered to design the IA.
- b. Briefly present the principles of Robustness.
- c. 'Recognition rather than recall' – Justify this heuristic with examples.

SECTION-B

Question 5 (Compulsory)

- a. Suppose you are planning for a tour next year, but can't choose among India, Singapore, Thailand, and Maldives. In case of going to any country from Bangladesh, you have options to choose a train instead of plane for India only. So the *transport* cost will reduce significantly if you go to India. Again, if you want to do *shopping*, Thailand as well as India will be appropriate whereas you wish for going to Maldives for years because you loved the *natural beauty* there. So with your limited budget you can *stay* for only 4 days in Singapore and Maldives. But for the other countries, you can stay at least 10 days. Moreover, in summer, India may not be a good option according to you.

Design a Graphical Issue Based Information System so that you can easily make the decision considering the pros and cons for each country.

- b. In windows, you can open the 'LinkedIn' app by pressing five keys: *shift, ctrl, alt, win* and *L*. You are planning to improve this by decreasing the number of keys pressed. For this, prepare the usability specification considering the conventional parameters for this task in windows.

Question 6

- a. Imagine a scenario:
In Fall 2022 semester, you registered for CSE 416 course. For registration, you read the course outline, contacted your advisor, and confirmed that you would take the course. After being assigned, you completed the payment. Now, in this course, you performed several submissions. Since it was a project based lab, you had to form a group, submit three project ideas, and finalize the project. To implement the project, you conducted a literature review as well as some usability evaluation studies following the methodology. At the end of course, you summarized the evaluation findings, compiled all data and writings as a report, and submitted via Google Classroom. If you wished, you also had a chance to go

your grade there.

Now construct a Graphical Hierarchy for completing CSE 416 course using Hierarchical Task Analysis (HTA).

- b. Consider two objects: one named "Assignment" which can be written, typed or not prepared' and can take 20 to 40 minutes to prepare. Another object is "Mary" who can perform two actions: prepare the assignment, attend classes and submit the assignment. Now write down the *relation* for the sentence "Mary wrote the assignment with pen" following Entity Relationship Techniques. 12

Question 7

- a. Imagine, you are browsing Gmail in a single user computer. You have already typed an email stating your thesis progress. To send the email, you will just type the email address of your supervisor and click the send button over there. And the mail is sent. Now, you will wait for a confirmation message. After successful sending of the email, a pop up containing confirmation message will be shown on the screen. Following the status event analysis approach, show the layers of elements along with their timelines and the interaction among them. Mark all the perceived events for the end user. 15+5=20
- b. Explain how Concur Task Trees are different from Hierarchical Task Analysis. 10

Question 8

- a. You are assigned to conduct an usability evaluation of specific models of vending machines. You have planned to conduct user study with at least 25 people who will perform the same tasks. Choose a *type of study*, *data collection method*, and *type of data analysis approach* with proper justification. 10
- b. Discuss how incremental prototyping can support iterative designs for interactive systems. 8
- c. What is Hypertext? Differentiate between linear text and hypertext by providing the necessary figures. 4+8=12

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SECTION-A

Question 1 (Compulsory)

- a. Illustrate the Information Architecture (IA) of an Education Management System (like Uniplex system of MIST) following the steps of Card Sorting Technique. At least 20 concepts need to be consider to design the IA. 16
- b. Explain the rationale behind selecting the card sorting technique for designing the system considered in question 1(a). 06
- c. Analyze the impact of order effects in within-subject design and explain how evaluators can mitigate the effects in their experimental design. 08

Question 2

- a. Evaluate any one of the following systems based on the seven universal design principles. Point out any omitted considerations on principles and suggest ways to enhance the system through redesign. 14
Choose a digital system for evaluation (any one):
 - i) Website of CSE department, MIST
 - ii) Wireless classroom microphone (the one used in your classroom)
 - iii) Slide changer (the one used in your classroom)
- b. "While playing a video game, experts tend to score less well when the sound is turned off compared to when it is on"- Explain the reasons behind the lower scores. 06
- c. Briefly differentiate between: 10
 - (i) Auditory icons vs. Earcons
 - (ii) Multimodal system vs. multi-media system.

Question 3

- a. Write short notes on: Scenario, Persona and Affordance 09
- b. State three characteristics of HCI design patterns 03
- c. Briefly discuss (with examples) the following usability principles introduced by prominent scholars: 18
 - i) Norman's 'Design for Error'
 - ii) Shneiderman's 'Easy reversal of Actions'

Question 4

- a. Discuss the differences and similarities between cognitive walkthrough and heuristic evaluation in the context of usability assessment.
- b. Outline the advantages and disadvantages associated with the 'Think Aloud' approach in usability testing.
- c. State the basic criterion for choosing an evaluation method.
- d. Justify the significance (whether it is significant or insignificant) of applying Miller's 7 ± 2 rule in designing navigation (menu size).

SECTION-B

Question 5 (Compulsory)

- a. State the properties of ISO Usability Standard 9241. Also, present a visualization of Throw-away prototyping within the requirement specification of Software Development Life Cycle.
- b. Consider a software company is assigned to develop a hall management system for MIST Osmany Hall. The hall provost has communicated with the responsible personnel of the company regarding the accommodation and mess management activities as features.
Provide an example of a possible case of formality gap in case of validation and requirement specifications.
- c. Suppose you are buying a phone case and a headphone from Daraz mobile app. You have added these two items to your cart already. Now, to check out, you have to tap on the cart icon, select your address, scroll down tapping twice to check products and bill respectively, then tapping on 'place order'. You are planning to improve the process by decreasing the number of taps. For this, prepare the usability specification considering the conventional parameters for this task.
Note: Provide necessary justification for current and your planned values.

Question 6

- a. Suppose, you are planning for furniture and appliance layout at your new home. You have the following products like round dining table, wall painting, bed, table lamp, wall calendar, dressing table, TV, wall clock, refrigerator, shoe rack, dining chair, hanging light, washing machine, router, dustbin, carpet and stool in your plan.

Build a Task Description hierarchy to organize these products according to relevancy and appropriateness considering the objects mentioned above.

- b. Consider three objects: One named 'class' which can be online, offline and canceled; and duration is 55 minutes. Another object is 'John' who can perform two actions: go to the classroom and attend HCI class. The last object is 'smart lock' which can be locked or unlocked; and is unlocked automatically. There is also an event called 'Level 04 (Term II)'.

Construct the following relations mentioning the type of relations

according to Entity relationship Technique:

- (i) Smart lock is established in the classroom.
- (ii) In level 04 (Term II), John attends HCI class.
- (iii) When John goes to the classroom, the smart lock is unlocked.
- (iv) John goes to the classroom on foot.

Question 7

- a. Imagine a scenario:

12+03

You are working on a file in Microsoft Word. At the same time, Google chrome is opened with your Gmail account logged in. Suddenly, you receive a notification sound for a new email. So, you press the mouse keeping your cursor on the chrome icon but in a hurry you move the cursor and release the button somewhere else.

Identify the next action performed by the OS. Following the Status Event Analysis approach, inspect the interactions among the layers of elements along with their timelines. Mark all the perceived events.

- b. Outline different types of triggers that are required to design a task or process. Provide one example for each type.
- c. Explain this statement – “Open loop control implies to a fragile system whereas for a robust system closed loop control is required.”

09

06

Question 8

- a. In case of analyzing data collected during user observation, differentiate between qualitative and quantitative data analysis with appropriate example.
- b. Model a design rationale using QOC notation for your facebook mobile app considering the options of access to the photos
(i) selected photos (ii) all photos (iii) none. Mention at least three criteria including “ease of use”.
- c. “While designing a website or software, there must be a proper balance between system complexity and multiple path available to the same page” – Share your opinion regarding this statement. Justify your opinion with appropriate example.

10

12

08

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SECTION-A

Question 1 (Compulsory)

- a. Find an appropriate evaluation method for the following session: 4+15
"you have a prototype for a theater booking system to be used by potential theater-goers to reduce queues at the box office."
Also, identify the followings:
 - i) The participants
 - ii) The technique used
 - iii) Representative tasks to be examined
 - iv) Measurements that would be appropriate
 - v) An outline plan for carrying out the evaluation
- b. In defining an experimental study, explain how would decide between a within-groups or between-groups experimental design with subject. 06
- c. Point out which Nielsen's heuristic is followed in the following design (notification given before sending the email). 05

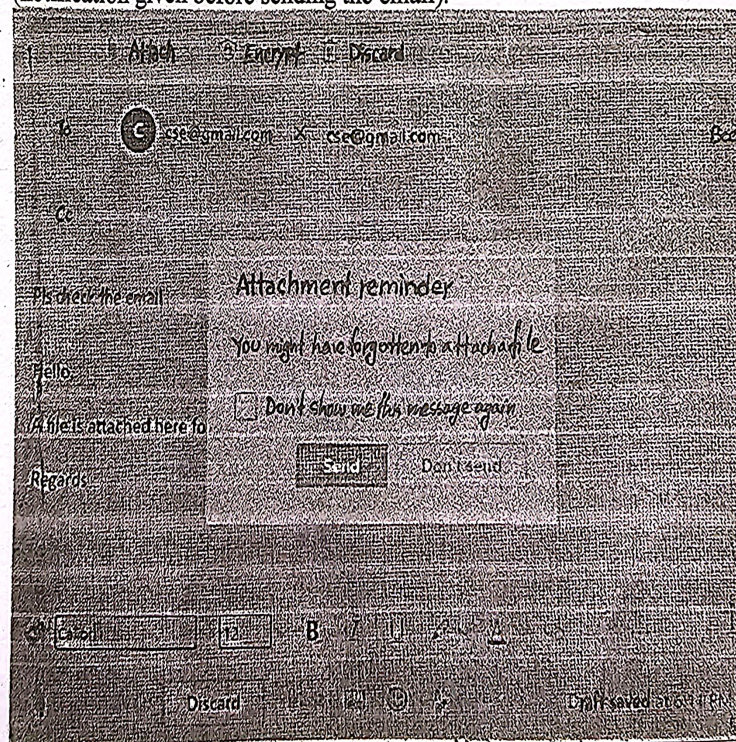


Figure 1(c): Notification given before sending email in an email application

Question 2

- a. Is multi-modality always a good thing? Justify your answer. 06
- b. What are (i) auditory icons and (ii) earcons? Describe their benefits to both visually impaired and sighted users. 10
- c. "A professor's website is designed so that it is accessible to everyone, including students who are blind and using text-to-speech software." 14
Evaluate the professor's website based on the seven general principles of universal design.

Question 3

- a. Define information architecture (IA). 04
- b. Depict the Information Architecture of an e-commerce website using the Card Sorting technique. At least 20 constructs need to be considered to develop the IA. 16
- c. 'Know the user' – Explain this statement in the context of Interaction design. 10

Question 4

- a. Briefly discuss the local structure and global structure of navigation design. 08
- b. Distinguish between principles, guidelines and standards, using examples of each to illustrate. 09
- c. Assess the 'MIST website' based on the five principles of Learnability. 13

SECTION-B**Question 5 (Compulsory)**

- a. Identify the differences between Persona and Scenario. Describe the importance of designing any software artifact. 12
- b. Briefly explain different types of prototyping. In case of each type, how can it be incorporated in the software life cycle? Show using necessary diagrams. 06+12

Question 6

Imagine a scenario:

You are sick and your doctor suggests you take some medicines. The medicines are kept on the dining table. It is very important to take medicine on time and you are missing your medicine often. So, you made a plan. You set a manual reminder from tonight for you. You first see the medicines on the table. Then, take the medicines and put them on your reading table. You are a student and so whenever you are sitting there, it will remind you about taking the medicines. Now, it's time to wait overnight for the next day. So, you brush your teeth, go to bed, sleep and wake up from bed in the next morning. Now, you are ready to study or we can say you will see the reminder set last night by you. So, you go to study room, sit on the chair and notice the medicine bottles kept on the table. You have different medicines to take at different times. So, you check the prescription, find the bottles which are prescribed to take before 30 minutes of meal, take the medicines, study for 30 minutes and eat

after meal medicines. When all the dosage are taken, you check all the bottles. For this, you take the bottle and check the medicine level, if it is empty then throw it to the bin.

- a. Now, construct a graphical Hierarchical Task Analysis (HTA) for the above mentioned scenario. 20
- b. What types of plans are included in your HTA? Mention the names with the cases as example of each type. (Just draw the specific plan as example) 10

Question 7

- a. Suppose you are a research assistant in a university. You are instructed to perform an analysis for your research on the interview taken of 20 participants. All the interviews are recorded on a disk. You are given the video recording and you have to extract the data for prespecified parameters, categorize them and perform the final analysis considering some criteria related to your research. 10

What type of analysis are you performing? Discuss the advantages of this type of analysis.

- b. Differentiate between Ethnography and participants observation. 10
- c. Write down short notes on the following: 05+05
 - i) Lost in Hyperspace
 - ii) Validation and Verification

Question 8

- a. Consider the scenario of receiving an image: 20

You are a user of WhatsApp. Message received in WhatsApp is stored in your mobile's internal memory. The sending and receiving of messages are managed by a tool-kit name 'WhatsApp Tool' on your phone. You see the messages on your screen. Now, you are reading a book online so attentively that you are not even looking anywhere else. Your friend John has sent you an image via WhatsApp. When the image is received, a notification popped up on the screen with sound. You tap it to see the image.

Now, show the time lines for various components in the interface using status-event analysis.

- b. Why HTML website also need to be user friendly from the point of view of HCI? Discuss with proper example. 10